Ukhanyiso

From matric marks to campus

See the courses you meet the minimum for, how close you are,
and the funding and documents you need to apply.

Ubuntu Innovators | FNB App of the Year Hackathon 2026 | Challenge: Access to higher education

Code: github.com/ConstanceKwena/ukhanyiso

VIDEO LINK: add before submitting

Team: Lorraine Mkwebo, Nhlanhla Mbokane, Kwena Mpherwane, Nana Lategang, Thembani Mahlulo

THE PROBLEM

A learner with their marks still can't tell where they can apply

345,000+

Bachelor passes in 2025, the most ever

34%

of 2025 candidates wrote Mathematics, which many courses require

Source: Department of Basic Education, 12 January 2026 (sanews.gov.za/node/79055; education.gov.za)

- Each institution publishes its own APS method, subject rules and levels.
- Closing dates are easy to miss. UP and TUT 2027 dates have already passed; UJ closes 31 October.
- Funding is a separate search, and every bursary asks for different certified documents.

WHO IT'S FOR

Built for the matriculant who is applying now

- Primary user: a recent matriculant or gap-year applicant, on a phone, reading their own results.
- Also: Grade 11 and 12 learners planning ahead, and parents or guardians helping with applications.
- Their questions: Which courses can I apply for? How close am I? What do I still need? Who can pay?

Mobile-first. No account needed to check courses. Demo login uses a synthetic profile only.

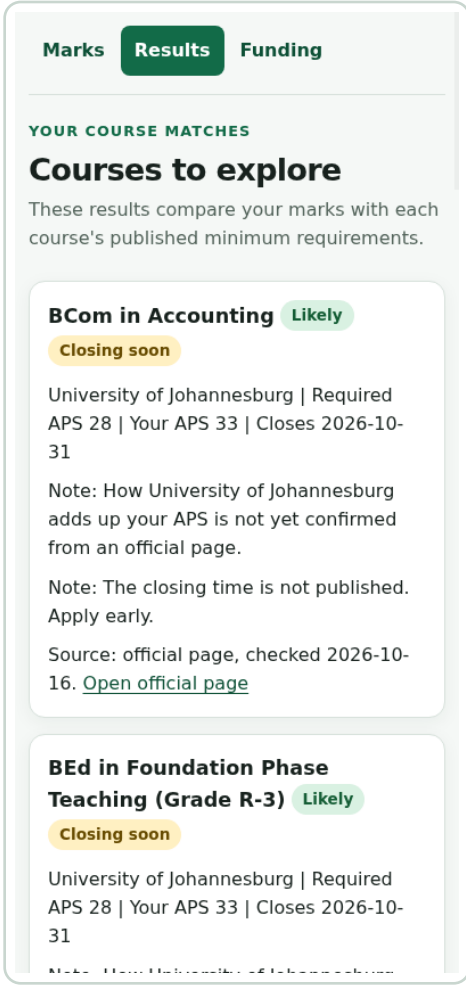
We do not store real learner names, ID numbers, marks or documents.

THE SOLUTION

Marks in, courses out

- Enter your marks. We convert them to levels and your APS the way each institution counts it.
- Each course gets a label: Likely, Possible or Stretch.
- A near-miss tells you exactly what closes the gap, for example "1 more APS point".
- Closing dates are shown separately, so "Applications closed" is never hidden.

We say "you meet the minimum requirements", never "you will get in".



Prototype screenshot with synthetic marks

HOW IT WORKS

Rule-based, sourced, and honest about gaps

1

Marks to levels

Each percentage becomes its national NSC level.

2

Levels to APS

Using each institution's published method.

3

Rules per course

Subjects, levels and APS, checked one by one.

4

Label or near-miss

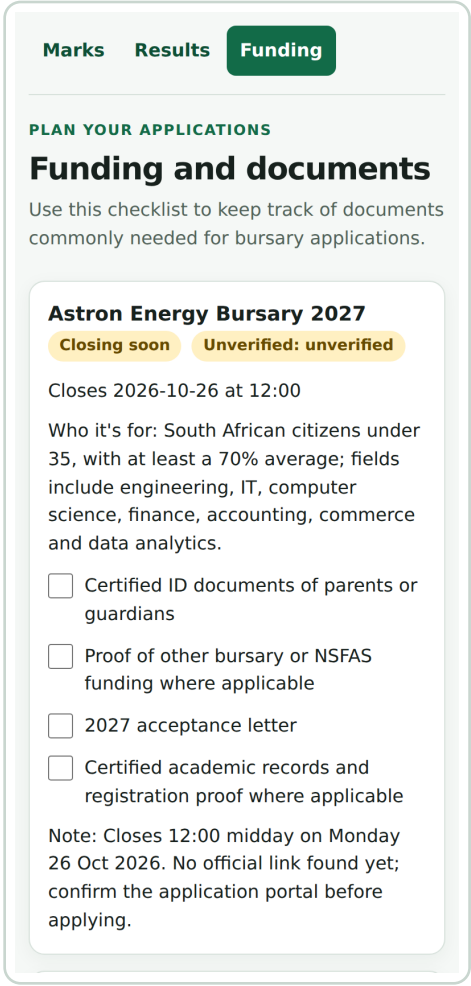
With a plain next step.

- Every course row records its source link, check date and status.
- Unverified data returns "cannot check" instead of a guess.
- Where an institution's APS method is not yet confirmed, results say so.
- Over 100 automated tests, including hand-calculated learner cases.

FUNDING CHECKLIST

Funding, and the documents you need

- Five 2027 bursaries and funds, including NSFAS, ISFAP and Funza Lushaka.
- Who each is for, the documents needed and the closing date.
- Each entry shows how well we have verified it, because sources sometimes disagree.
- A checklist API with a demo login (synthetic account) is built; the page does not save ticks yet.



Prototype screenshot, 22 Oct 2026

VALIDATION

User testing is planned, results to follow

We have not yet run user tests, so we do not quote results.

- Planned for Sunday 25 October with 3 to 5 recent matriculants.
- Five tasks: find your courses, read a label, act on a near-miss, spot a closed course, find a bursary.
- Consent first, no names recorded, made-up or own marks, the learner can stop any time.

Our targets

At least 4 of 5 find their courses unaided, explain the labels, and understand a near-miss.

Nobody should be confused by "Applications closed".

SECURITY AND ROADMAP

Safe by design, and what comes next

Built now

- Match requests limited to 10 per minute.
- Synthetic data only, no real learner data stored.
- Demo login: hashed passwords, CSRF protection, session replaced on login.
- Demo runs locally; data resets on restart (in-memory H2).

Not built yet (roadmap)

- More courses and institutions, including TVET colleges.
- First-year survival hub.
- WhatsApp bot.
- Security checks still to run: secrets scan, dependency alerts, secret scanning, security testing.

Always confirm requirements and dates with the institution before applying.

Course data is from official pages, checked 16 October 2026.